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suffi@PC: ~/xv6-labs-2025
riscv64-linux-gnu-objdump -S user/_uptime > user/uptime.asm
riscv64-linux-gnu-objdump -t user/_uptime | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/uptime.sym
riscv64-linux-gnu-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mcmodel=medany
-fffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset
-fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-
builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprint
f -fno-builtin-vprintf -l. -fno-stack-protector -fno-pie -no-pie -c -o user/memdump.o user/memdump.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_memdump user/memdump.o user/ulib.o user/usys.o user/
printf.o user/umalloc.o
riscv64-linux-gnu-objdump -S user/_memdump > user/memdump.asm
riscv64-linux-gnu-objdump -t user/_memdump | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/memdump.sym
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user
/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie
user/_sleep user/_logstress user/_forphan user/_dorphan user/_find user/_sixfive user/_uptime user/_memdump user/_find
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1143 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-l
egacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ uptime
ticks: 28
$ uptime
ticks: 2066
$ |

```

2

```

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riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_memdump user/memdump.o user/ulib.o user/usys.o user/
printf.o user/umalloc.o
riscv64-linux-gnu-objdump -S user/_memdump > user/memdump.asm
riscv64-linux-gnu-objdump -t user/_memdump | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/memdump.sym
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user
/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie
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egacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ find . f$
$ find . ^f
./findtest.sh
./forktest
./forphan
./find
./find
$ find . a.
./cat
./forphan
./dorphan
$

```

3

suffi@PC: ~/xv6-labs-2025

```
init: starting sh
sleep 10
wait
echo done
done
echo first
first
echo second
second
history
1: sleep 10

2: wait

3: echo done

4: echo first

5: echo second

complete ec
Completions for 'ec':
  echo
!!
Completions for 'ec':
  echo
  echo hi
  hi
  history
1: sleep 10

2: wait

3: echo done
```